GENESIS
IDEATHON

TEAM & DOMAIN

- Team Name – Vortex
- Team Members –
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- Selected Domain – 6
- Problem Statement –Heart and lung diseases are increasingly affecting the growing population. These organs are checked separately through complex, expensive or one time use equipments. However, early detection remains critical but unachieved in rural communities and among the younger generation, leaving critical early-stage conditions undiagnosed.



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DOMAIN PROBLEM IN OUR WORDS

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What exactly needs to be solved?

Today, smoke exposure, lung function and heart-related signals are typically checked separately, using different devices and workflows. These tools can be expensive, clinic dependent and hard to access in rural and community settings. What is needed is a simple, cheap screen to bring these signals together and tell you exactly what is behind a concerning report.

Why is it important right now?

Both smoking and household biomass smoke affect large populations and early warning signs may be missed until symptoms develop. There is a need for rural communities, schools and community health programs to have simple means of identifying those who may require further medical investigation.
The second wave of COVID-19 showed the importance of rapid community screening, pulse oximeters identified “silent hypoxia” but an oximeter alone cannot determine airflow limitation or smoke exposure.

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WHO IS AFFECTED?

Our target group

ASHA (Accredited Social Health Activist) and ANM (Auxiliary Nurse Midwife) who conduct doorstep visits; PHC (Primary Health Care); high-exposure individuals: smokers and households using biomass for cooking, who are at risk for early heart and lung disease.



Secondary stakeholders

Occupational and ngo field teams.Schools, colleges- for STEM education and non-clinic awareness.

Who benefits if the problem is solved

Our main target is the healthcare system in general, and to be particular, we are targeting people living in rural and underserved communities where they can receive combined heart and lung early warnings before severe symptoms develop.

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CURRENT SITUATION & CONTEXT

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Existing scenario and background

Lung and heart diseases are linked to harmful smoke exposure, yet health screenings are performed separately in specialized clinics using a single purpose or expensive equipment.

Schools, rural health workers and community health programs lack low-cost, portable tool which would help detect early warning signs before its too late.

Relevant data or facts

267 million adults in India use tobacco

199 million of these are from rural areas

Over 40% of households mainly use solid biomass fuel

Cardiovascular disease accounts for 28.1% of all deaths

Current workflow: Step by Step

Smoke Exposure
Smokerlyzer

↓

Breath flow
Spirometer

↓

Heart/Oxygen
Pulse oximeter/ECG

↓

Separate results
Different devices

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KEY CHALLENGES

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Main difficulties and bottlenecks

Biomass smoke can be misinterpreted. Exhaled CO can come from tobacco or biomass combustion, so a rural cook should not automatically be labelled a smoker.
Many clinical devices require trained personnel and controlled settings.

Rural and community settings are limited and may not have access to specialized, expensive screening equipment.

Root causes behind them

Usually, healthcare treats respiratory or cardiac problems as completely separate clinical domains rather than linked matters.

Standard smoking tools miss solid biomass fuel smoke, which affects over 40% of Indian households (disproportionately rural women and children)

Consequences of leaving them unsolved

Potential warning signs may remain unnoticed until symptoms appear, while health workers lack a simple first-line tool for deciding who needs counselling, rechecking or referral.

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EXISTING SOLUTIONS / APPROACHES

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Feature	Smokerlyzer	Clinical spirometer	Pulse oximeter	Smartwatch	Heart-Lung Check
CO smoke marker	Yes	No	No	No	Yes
Breath flow	No	Yes	No	No	Screening-grade (peak flow + 1-s volume index)
SpO_2 and heart rate	No	No	Yes	Yes	Yes
Single-lead ECG	No	No	No	Some	Yes (rhythm flag only)
One explainable score	No	No	No	No	Yes (Green/Yellow/Red + driver)
Biomass-context flag	No	No	No	No	Yes
Offline rural use	Partial	Yes	Yes	No	Yes
Indicative price	₹40k–70k	₹50k–1.5L	₹1k–3k	₹15k–40k	₹14,500 (BOM ₹9,000)

THE GAP:
No existing option in the comparison combines smoke exposure, breath flow and heart/oxygen signals into one explainable screening workflow designed for low-cost community use.

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PROBLEM INSIGHTS & OPPORTUNITY AREAS

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HARDWARE REQUIREMENTS

- **Controller** - ESP32
- **Cardiac + oxygen** - MAX86150
  - Fallback: MAX30102 + AD8232
- **Exhaled CO** - ZE07-CO
  - Fallback: ME3-CO / 3-electrode CO cell + LMP91000
- **Breath flow** - SDP810-500Pa + 3D-printed Venturi
- **Environmental correction** - BME280
- **User interface** - OLED / display
- **Power** - Rechargeable Li-ion battery
- **Hygiene** - Single-use mouthpiece + hydrophobic filter + one-way valve

SOFTWARE REQUIREMENTS

- ESP32 firmware
- Sensor data acquisition
- Signal-quality checks
- Environmental correction
- Screening-rule engine
- Green / Yellow / Red result generation
- Driver identification
- Local data storage
- Optional consent-based cloud sync
- Dashboard / visualisatio

OTHER RESOURCES

- Clinical equipments
- Calibration equipment
- ASHA/ANM and teacher usability testing

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INITIAL DIRECTION & EXPECTED OUTCOME

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WHAT WE INTEND TO DO

Develop a guided, portable screening station that measures exhaled CO, breath flow, heart rate, SpO_2 and single-lead ECG in approximately one minute of active testing.

Channel	Weight*	Green	Yellow	Red
Breath flow	40%	≥ 80% of predicted	60–79% of predicted	< 60% of predicted
Exhaled CO (ambient-corrected)	30%	0–6 ppm	7–9 ppm	≥ 10 ppm
SpO_2	20%	≥ 95%	90–94%	< 90%
Cardiac signal	10%	60–100 bpm, regular rhythm	50–59 or 101–110 bpm, or occasional irregular beats	< 50 or > 110 bpm, or sustained irregular rhythm

WHAT WE WANT TO IMPROVE

- **Affordability**
- **Accessibility**
- **Clarity**

SUCCESS SHOULD MEAN

- Valid screening completed in under 2 minutes
- ≥90% of users complete a valid test within two attempts
- Median screening time <2 minutes
- Sensor accuracy validated against reference devices
- Clear explanation of the factor driving the result
- Appropriate counselling/re-referral pathway